Chapter 5, Problem 35P

Problem

Design a noninverting amplifier with a gain of 7.5.

Step-by-step solution

Step 1 of 3

Draw the non-inverting amplifier circuit diagram as shown in Figure 1.

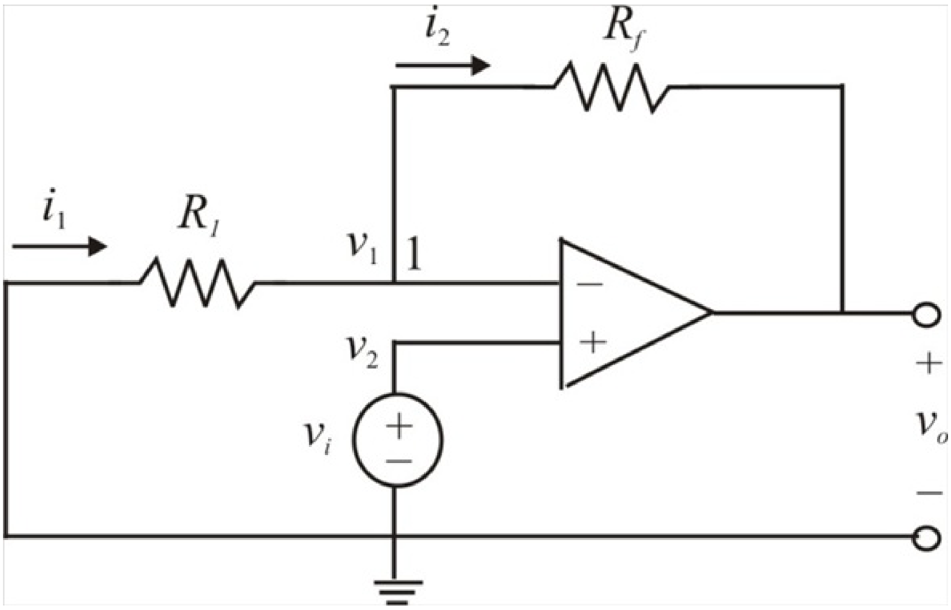

Figure 1

Step 2 of 3

The gain of the amplifier $A_v = 7.5$

Applying Kirchhoff's current law to node 1,

$$i_1 = i_2$$

$$\frac{0 - v_1}{R_1} = \frac{v_1 - v_o}{R_f}$$

But, $v_1 = v_2 = v_i$ for an ideal op amp

Substituting $v_1 = v_i$ in the above expression, we get

$$\frac{-v_i}{R_1} = \frac{v_i - v_o}{R_f}$$

$$v_o = \left(1 + \frac{R_f}{R_1}\right) v_i$$

Step 3 of 3

The voltage gain is

$$A_v = \frac{v_o}{v_i} = 1 + \frac{R_f}{R_1}$$

Therefore we have,

$$1 + \frac{R_f}{R_1} = 7.5$$

$$\frac{R_f}{R_1} = 6.5$$

$$R_f = 6.5\,R_1$$

If the resistance $R_1 = 60\,\text{k}\Omega$, then the feedback resistance is

$$\begin{aligned} R_f &= 6.5\,R_1 \\ &= 6.5(60\,\text{k}\Omega) \\ &= 390\,\text{k}\Omega \end{aligned}$$

Therefore the resistance R_1 is $\boxed{60\,\text{k}\Omega}$.

Therefore the feedback resistance R_f is $\boxed{390\,\text{k}\Omega}$.